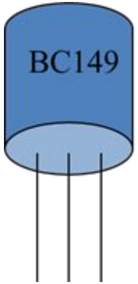
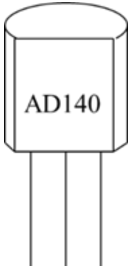
Transistor

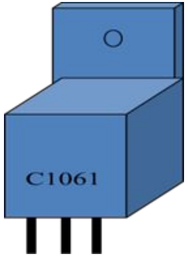
- Invented in 1948 by J. Bardeen and W.H. Brattain of Bell Telephone Laboratories, U.S.A.;
- Transistor has now become the heart of most electronic applications.



(i) BC149 (NPN- Si)



(ii) AD140 (PNP- Ge)



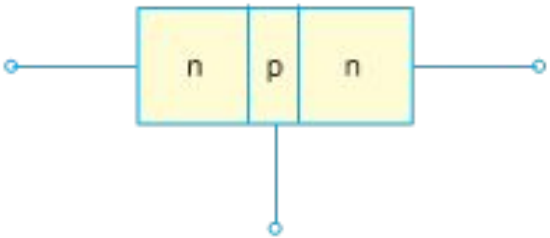
(iii) C1016 (NPN- Si)

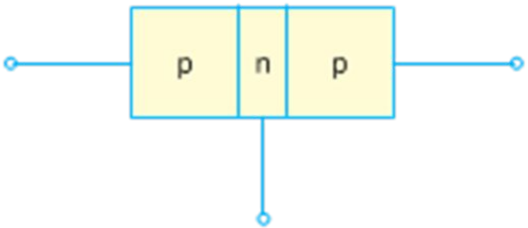
Transistor

A transistor consists of two pn junctions formed by sandwiching either p-type or n-type semicon- ductor between a pair of opposite types. Accordingly ; there are two types of transistors, namely;

(i) n-p-n transistor

(ii) p-n-p transistor





- These are two pn junctions. Therefore, a transistor may be regarded as a combination of two diodes connected back to back.
- There are three terminals, one taken from each type of semiconductor.
- The middle section is a very thin layer. This is the most important factor in the function of a transistor.

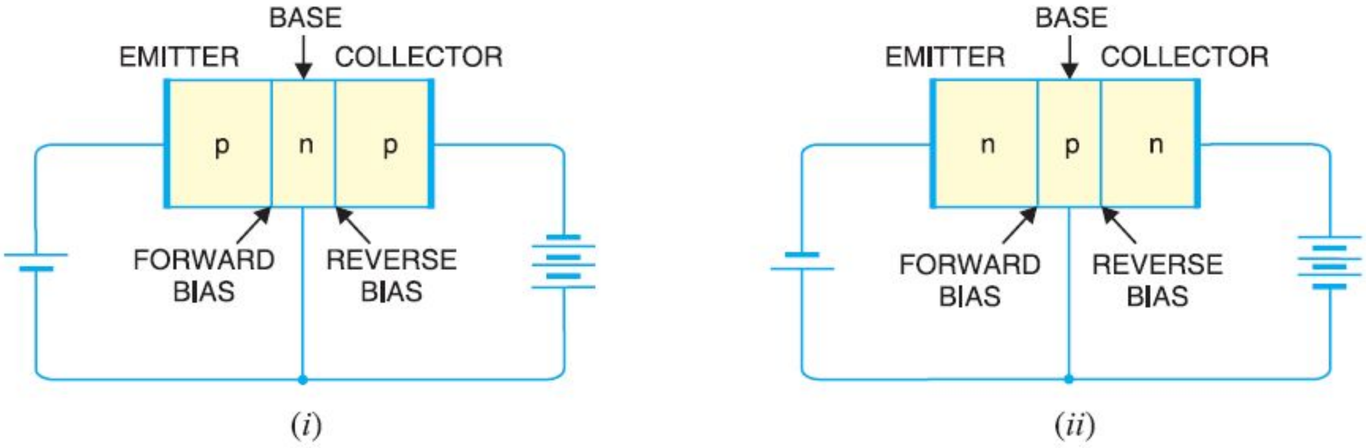
Origin of the name "Transistor"

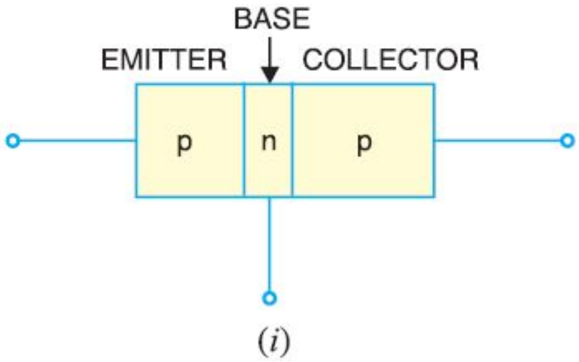
- A transistor has two pn junctions.
- One junction is forward biased and the other is reverse biased.
- The forward biased junction has a low resistance path whereas a reverse biased junction has a high resistance path.
- The weak signal is introduced in the low resistance circuit and output is taken from the high resistance circuit.
- Therefore, a transistor transfers a signal from a low resistance to high resistance.
- The prefix 'trans' means the signal transfer property of the device while 'istor' classifies it as a solid element in the same general family with resistors.

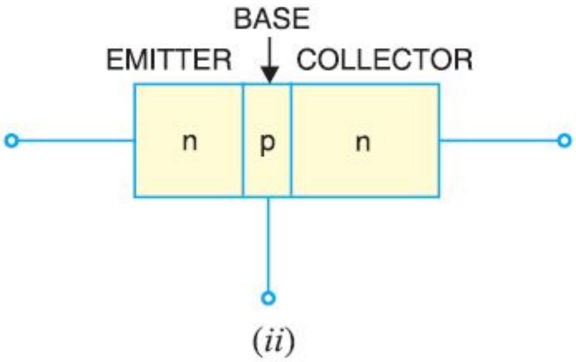
Transfer + Resistor → Transistor

Naming the Transistor Terminals

- Emitter
- Collector
- Base

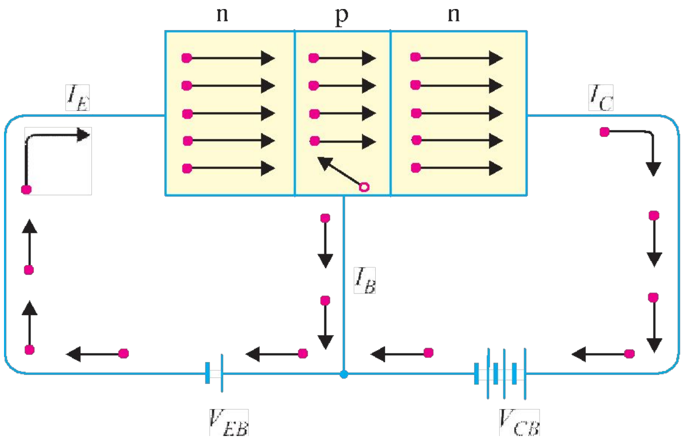






- The transistor has three regions, namely ; emitter, base and collector.
- The base is much thinner than the emitter while collector is wider than both.
- The emitter is heavily doped so that it can inject a large number of charge carriers (electrons or holes) into the base.
- The base is lightly doped and very thin ; it passes most of the emitter injected charge carriers to the collector.
- The collector is moderately doped.

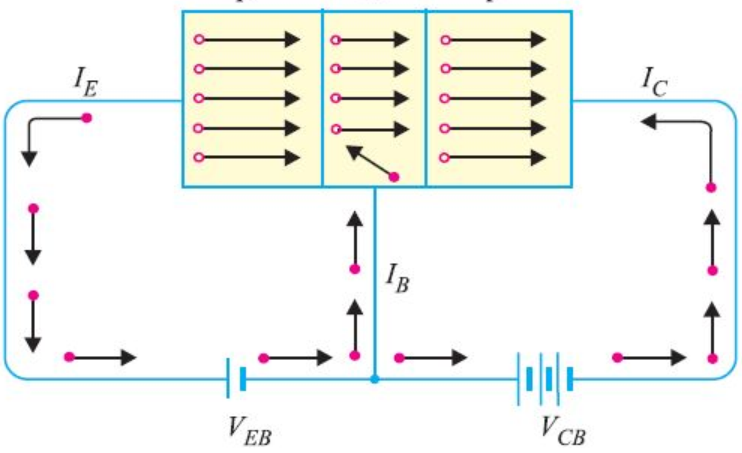
Transistor Action



$$I_E = I_B + I_C$$

- Electrons of n-type flow towards the base that constitutes the emitter current I_E.
- less than 5% combine with holes to constitute base current I_B.
- more than 95%) cross over into the collector region to constitute collector current I_C.

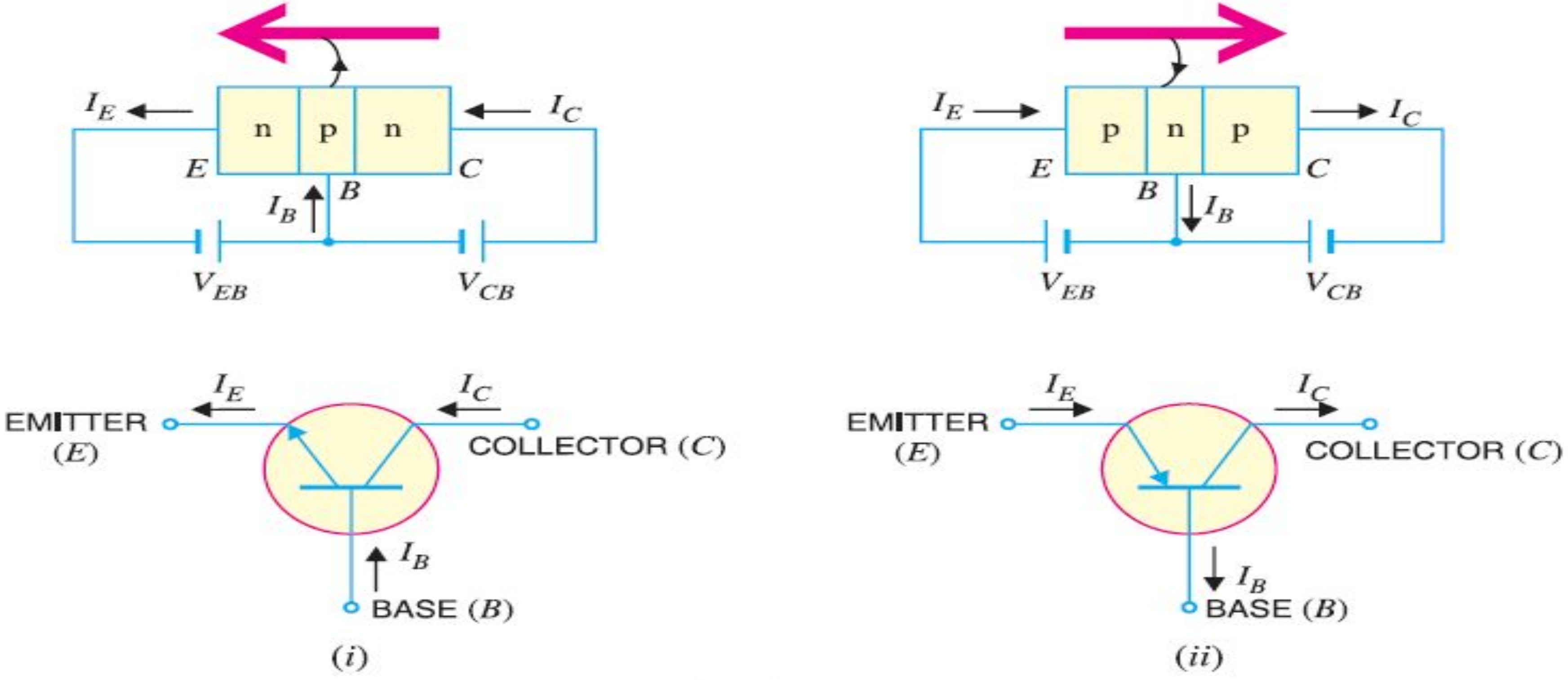
$I_E = I_B + I_C$



Basic connection of *pnp* transistor

- Holes of p-type flow towards the base that constitutes the emitter current I_E.
- less than 5% combine with electrons to constitute base current I_B.
- more than 95% cross over into the collector region to constitute collector current I_C.

Transistor Symbols

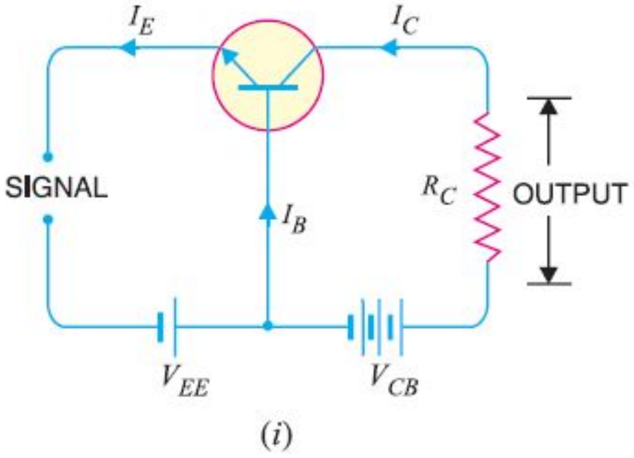


Transistor Connections

Transistor can be connected in a circuit in the following three ways :

1. Common Base Connection
2. Common Emitter Connection
3. Common Collector Connection

Common Base Connection



Common base npn transistor   Common base pnp transistor

Current amplification factor (α): *The ratio of change in collector current to the change in emitter current at constant collector- base voltage V_{CB}.*

range from 0.9 to 0.99

$$\alpha = \frac{\Delta I_C}{\Delta I_E}$$

$$For\ \text{d.c.},\ \alpha = \frac{I_C}{I_E}$$

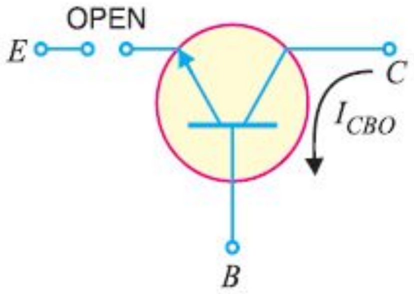
Expression for collector current:

Total collector current, $I_C = \alpha I_E + I_{leakage}$

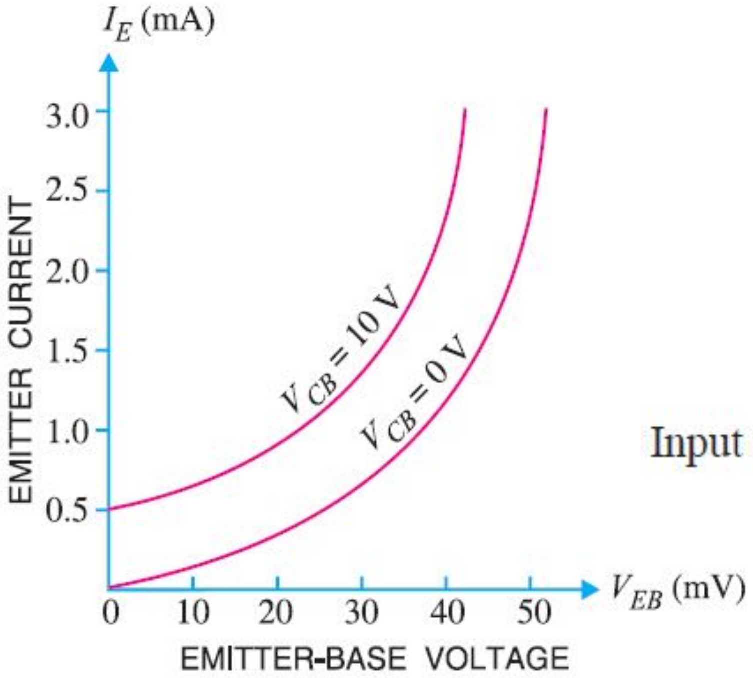
If $I_E = 0$, a small leakage current still flows in the collector circuit.

So, $I_{leakage} = I_{CBO}$, meaning collector-base current with emitter open.

$$
\begin{aligned}
I_C &= \alpha I_E + I_{CBO} \\
I_E &= I_C + I_B \\
I_C &= \alpha (I_C + I_B) + I_{CBO} \\
I_C (1 - \alpha) &= \alpha I_B + I_{CBO} \\
I_C &= \frac{\alpha}{1 - \alpha} I_B + \frac{I_{CBO}}{1 - \alpha}
\end{aligned}
$$



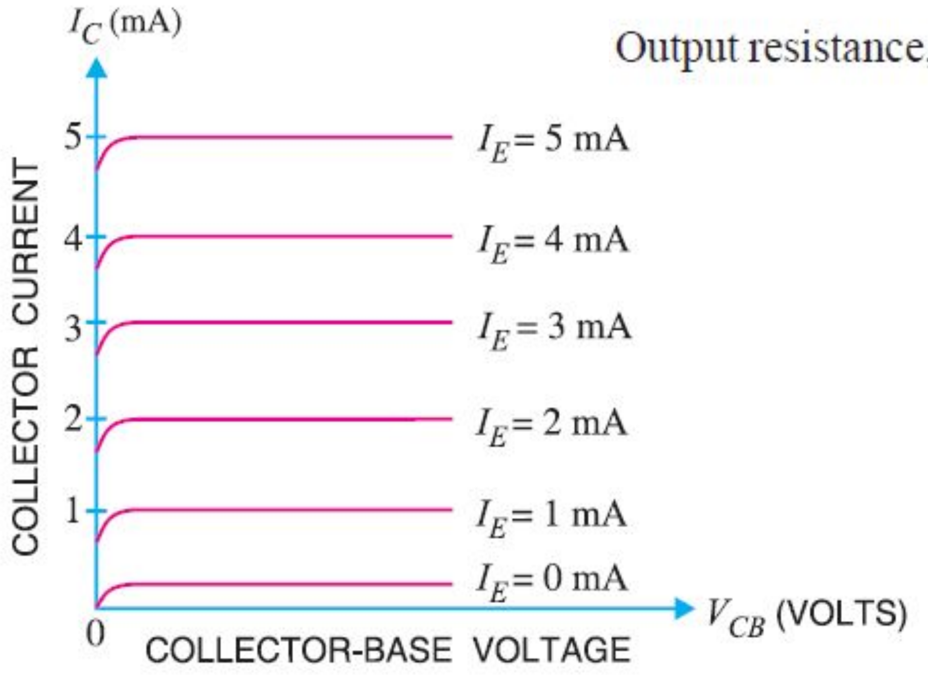
Input characteristic:



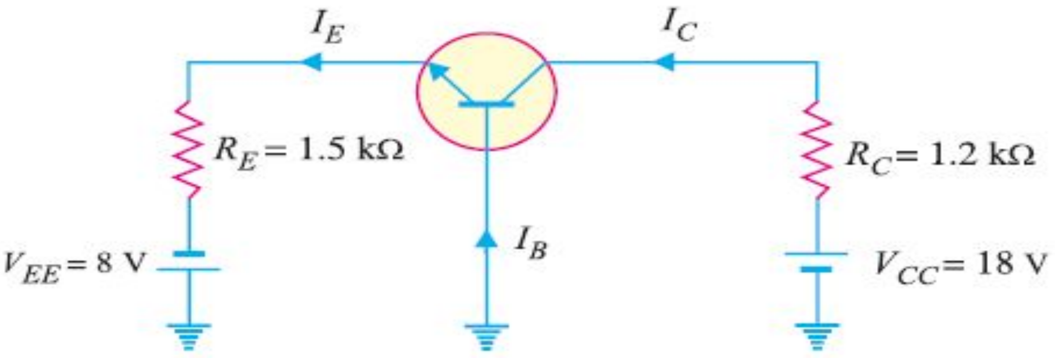
Input resistance, $r_i = \dfrac{\Delta V_{BE}}{\Delta I_E}$ at constant V_{CB}

Output characteristic:

Output resistance, $r_o = \dfrac{\Delta V_{CB}}{\Delta I_C}$ at constant I_E



Example 8.7. *For the common base circuit shown in Fig., determine IC and VCB. Assume the transistor to be of silicon.*



$$V_{EE} = I_E R_E + V_{BE}$$

$$I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

$$= \frac{8V - 0.7V}{1.5\ k\Omega} = 4.87\ \text{mA}$$

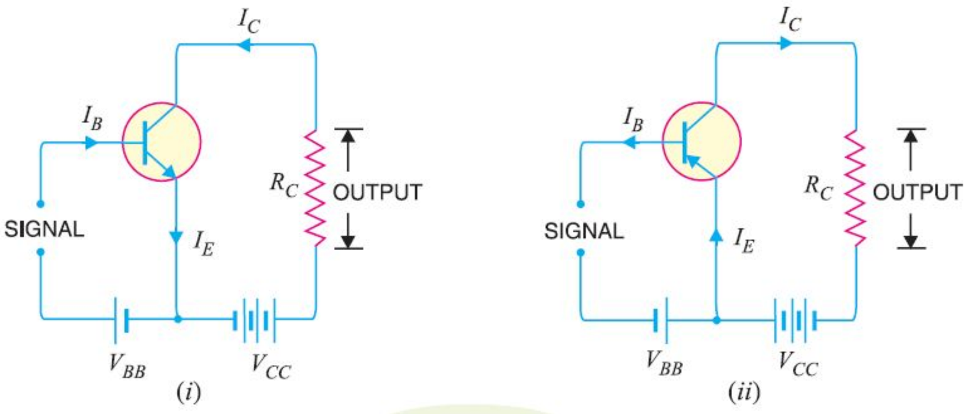
$$\therefore \quad I_C \simeq I_E = \mathbf{4.87\ mA}$$

$$V_{CC} = I_C R_C + V_{CB}$$

$$V_{CB} = V_{CC} - I_C R_C$$

$$= 18\ \text{V} - 4.87\ \text{mA} \times 1.2\ k\Omega = \mathbf{12.16\ V}$$

Common Emitter Connection



Base current amplification factor (β):

ranges from 20 to 500

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

$$For\ \text{d.c.},\ \beta = \frac{I_C}{I_B}$$

Relation between β and α:

$$\beta = \frac{\Delta I_C}{\Delta I_B} \qquad ...(i)$$

$$\alpha = \frac{\Delta I_C}{\Delta I_E}$$

$$I_E = I_B + I_C$$

$$\Delta I_E = \Delta I_B + \Delta I_C$$

$$\Delta I_B = \Delta I_E - \Delta I_C$$

$$\beta = \frac{\Delta I_C / \Delta I_E}{\dfrac{\Delta I_E}{\Delta I_E} - \dfrac{\Delta I_C}{\Delta I_E}} = \frac{\alpha}{1-\alpha}$$

$$\beta = \frac{\alpha}{1-\alpha}$$

Substituting the value of ΔI_B in exp. (*i*), we get,

$$\beta = \frac{\Delta I_C}{\Delta I_E - \Delta I_C} \qquad ...(iii)$$

Dividing the numerator and denominator of R.H.S. of exp. (*iii*) by ΔI_E, we get,

- It is clear that as α approaches unity, β approaches infinity.
- In other words, the current gain in common emitter connection is very high.
- It is due to this reason that this circuit arrangement is used in about 90 to 95 percent of all transistor applications.

Expression for collector current:

We know $I_E = I_B + I_C$

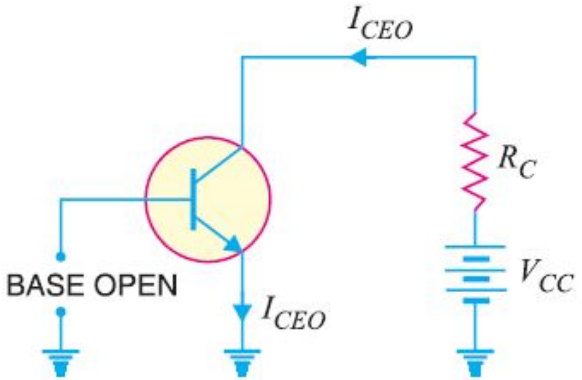
$I_C = \alpha I_E + I_{CBO}$

From exp. (*ii*), we get,

$$I_C = \alpha I_E + I_{CBO} = \alpha (I_B + I_C) + I_{CBO}$$

$$I_C (1 - \alpha) = \alpha I_B + I_{CBO}$$

$$I_C = \frac{\alpha}{1-\alpha} I_B + \frac{1}{1-\alpha} I_{CBO} \quad \ldots(iii)$$



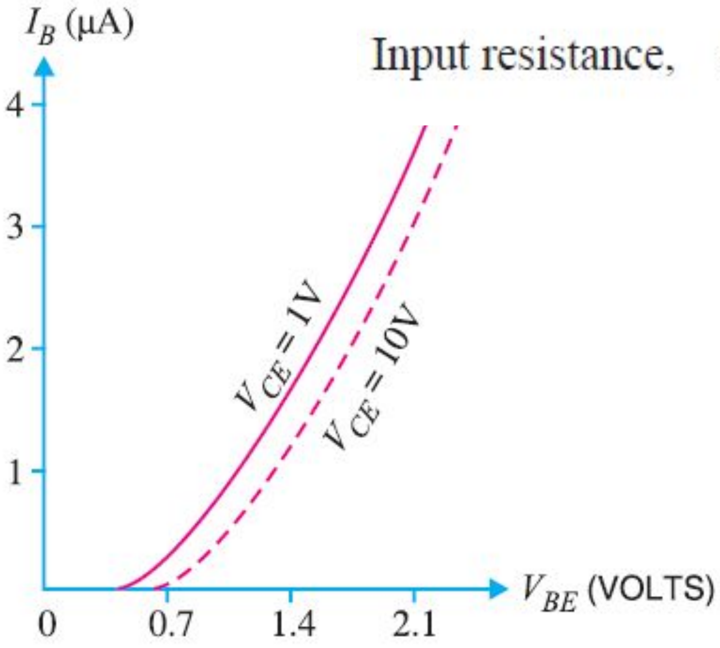
From exp. (*iii*), it is apparent that if $I_B = 0$

$$I_{CEO} = \frac{1}{1-\alpha} I_{CBO}$$

$$I_C = \frac{\alpha}{1-\alpha} I_B + I_{CEO}$$

$$I_C = \beta I_B + I_{CEO}$$

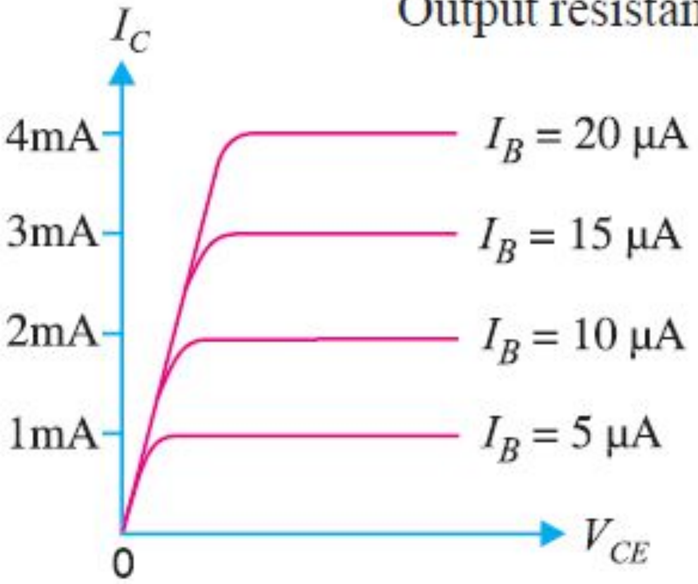
Input characteristic:



Input resistance, $r_i = \dfrac{\Delta V_{BE}}{\Delta I_B}$ at constant V_{CE}

Output characteristic:

Output resistance, $r_o = \frac{\Delta V_{CE}}{\Delta I_C}$ at constant I_B



Example 8.12. *If α = 0.96, determine : collector-emitter voltage, base current.*

(*i*) Collector-emitter voltage,

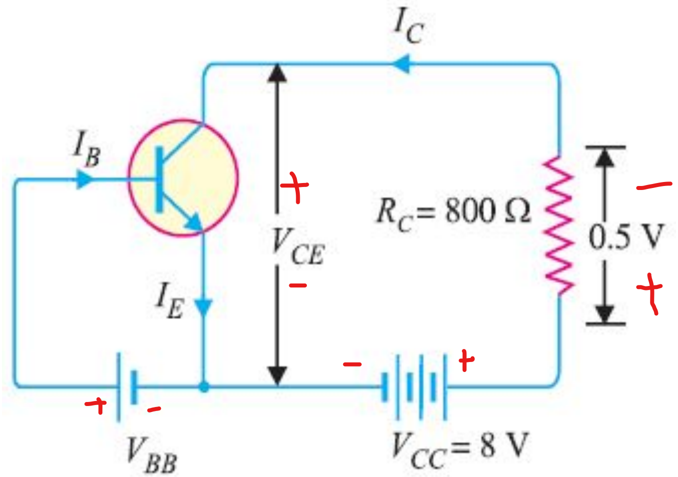
$$V_{CE} = V_{CC} - 0.5 = 8 - 0.5 = \mathbf{7.5\ V}$$

(*ii*) The voltage drop across R_C (= 800 Ω) is 0.5 V.

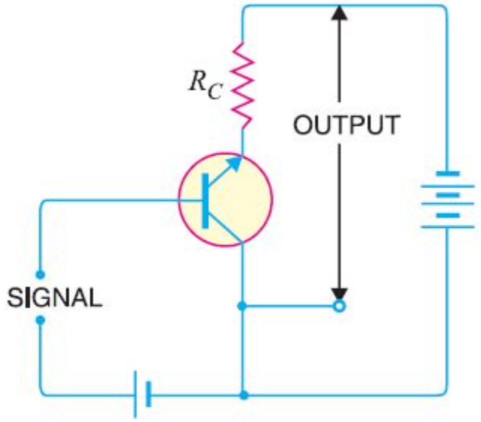
$$\therefore \quad I_C = \frac{0.5\ \text{V}}{800\ \Omega} = \frac{5}{8}\ \text{mA} = 0.625\ \text{mA}$$

$$\text{Now} \quad \beta = \frac{\alpha}{1-\alpha} = \frac{0.96}{1-0.96} = 24$$

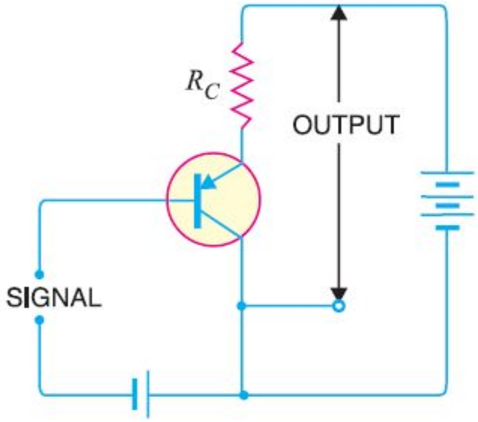
$$\text{Base current,} \quad I_B = \frac{I_C}{\beta} = \frac{0.625}{24} = \mathbf{0.026\ mA}$$



Common Collector Connection



(*i*)



(*ii*)

Current amplification factor γ:

$$\gamma = \frac{\Delta I_E}{\Delta I_B}$$

Its current gain is same as common emitter circuit.

Relation between γ and α:

$$\gamma = \frac{\Delta I_E}{\Delta I_B} \qquad \ldots(i)$$

Dividing the numerator and denominator of R.H.S. by ΔI_E, we get,

$$\gamma = \frac{\dfrac{\Delta I_E}{\Delta I_E}}{\dfrac{\Delta I_E}{\Delta I_E} - \dfrac{\Delta I_C}{\Delta I_E}} = \frac{1}{1-\alpha}$$

$$\therefore \qquad \gamma = \frac{1}{1-\alpha}$$

Substituting the value of ΔI_B in exp. (i), we get,

$$\gamma = \frac{\Delta I_E}{\Delta I_E - \Delta I_C}$$

Expression for collector current:

$$I_C = \alpha I_E + I_{CBO}$$

$$I_E = I_B + I_C = I_B + (\alpha I_E + I_{CBO})$$

$$I_E(1-\alpha) = I_B + I_{CBO}$$

$$I_E = \frac{I_B}{1-\alpha} + \frac{I_{CBO}}{1-\alpha}$$

$$I_C \; ; \; I_E = (\beta+1) I_B + (\beta+1) I_{CBO}$$

Applications:

- very high input resistance (about 750kΩ) and very low output resistance (about 25Ω).
- seldom used for amplification.

Comparison of Transistor Connections

S. No.	Characteristic	Common base	Common emitter	Common collector
1.	Input resistance	Low (about 100 Ω)	Low (about 750 Ω)	Very high (about 750 kΩ)
2.	Output resistance	Very high (about 450 kΩ)	High (about 45 kΩ)	Low (about 50 Ω)
3.	Voltage gain	about 150	about 500	less than 1
4.	Applications	For high frequency applications	For audio frequency applications	For impedance matching
5.	Current gain	No (less than 1)	High (β)	Appreciable

Commonly Used Transistor Connection

The common emitter circuit is used in about 90 to 95 per cent of all transistor applications. The main reasons are:

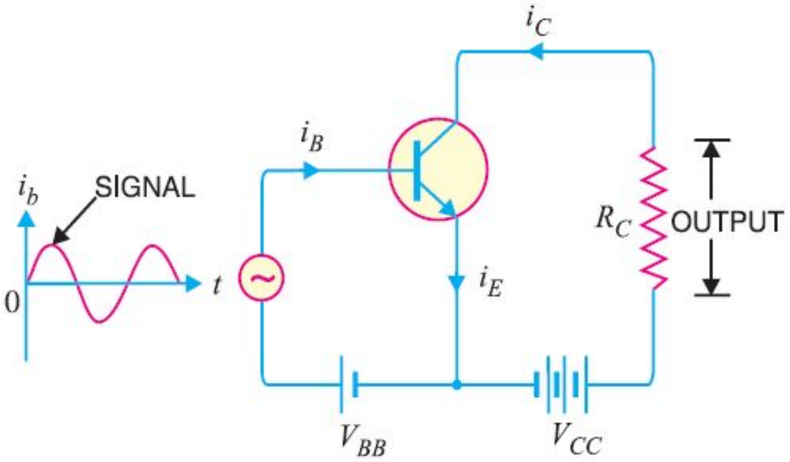
- High current gain

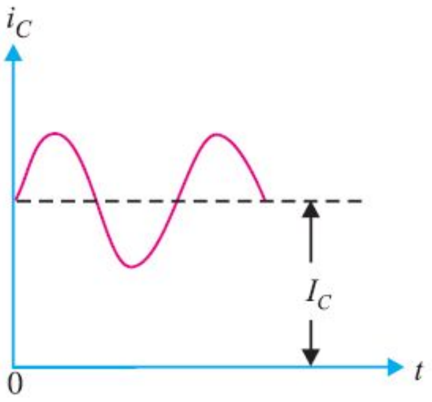
$$I_C = \beta I_B + I_{CEO}$$

It may range from 20 to 500

- High voltage and power gain
- Moderate output to input impedance ratio

Transistor as an Amplifier in CE Arrangement





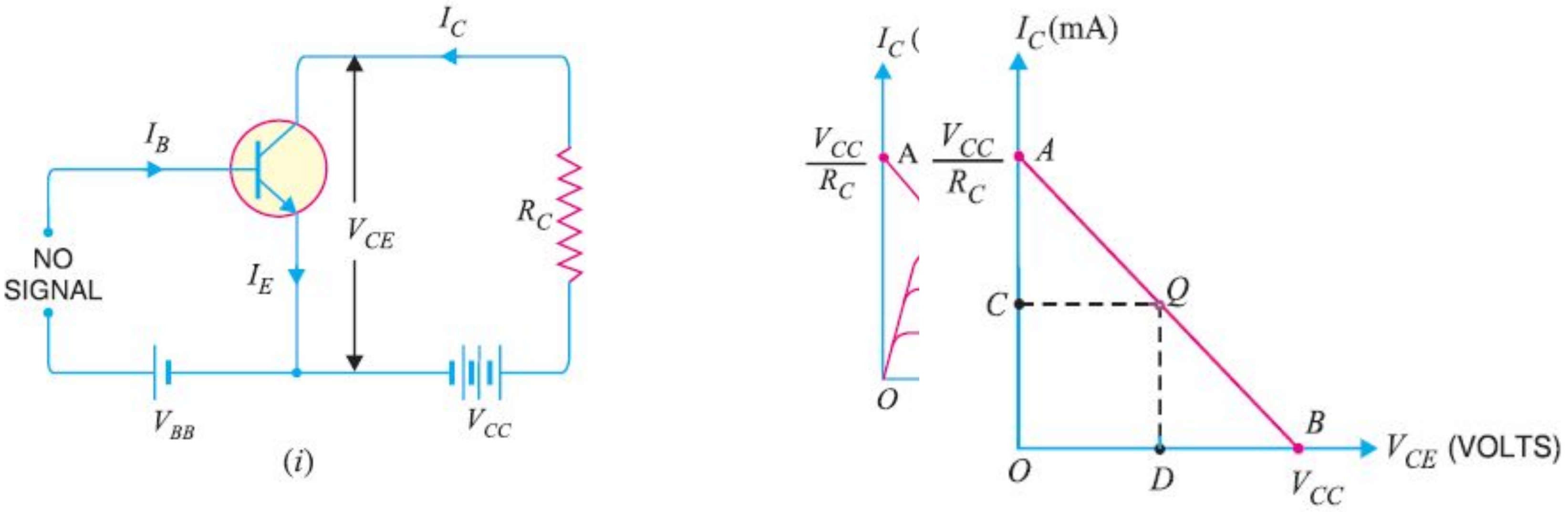
- I_C (zero signal collector current) due to bias battery V_{BB}.
- a.c. collector current i_c due to signal.

∴ Total collector current, $i_C = i_c + I_C$

Transistor Load Line Analysis

d.c. load line:

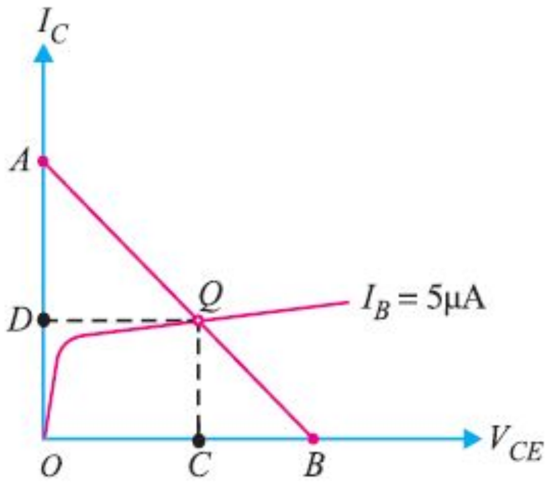
$$V_{CE} = V_{CC} - I_C R_C$$



For any value of collector-current OC, the collector-emitter voltage is OD.

Operating Point

- The zero signal values of Ic and V_{CE} are known as the operating point.
- It is so called because the variations of Ic and V_{CE} take place about this point when the signal is applied.
- Also called quiescent (silent) point or Q-point because it is the point on Ic-V_{CE} characteristic when the transistor is silent i.e. in the absence of the signal.

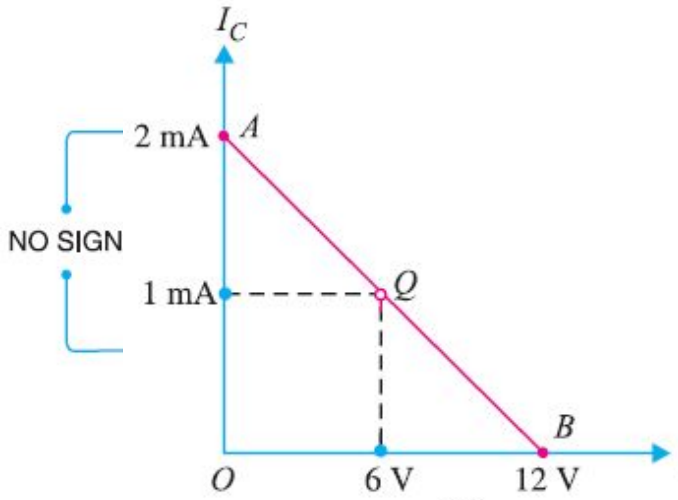


Example 8.23. *In the circuit diagram shown in Fig. 8.39 (i), if $V_{CC} = 12V$ and $R_C = 6\ k\Omega$, draw the d.c. load line. What will be the Q point if zero signal base current is 20μA and $\beta = 50$?*

$$V_{CE} = V_{CC} - I_C R_C$$

When $I_C = 0$, $V_{CE} = V_{CC} = 12$ V

$I_C = V_{CC}/R_C = 12\ \text{V}/6\ \text{k}\Omega = 2$ mA.



(ii)

Zero signal base current, $I_B = 20\ \mu\text{A} = 0.02$ mA

Current amplification factor, $\beta = 50$

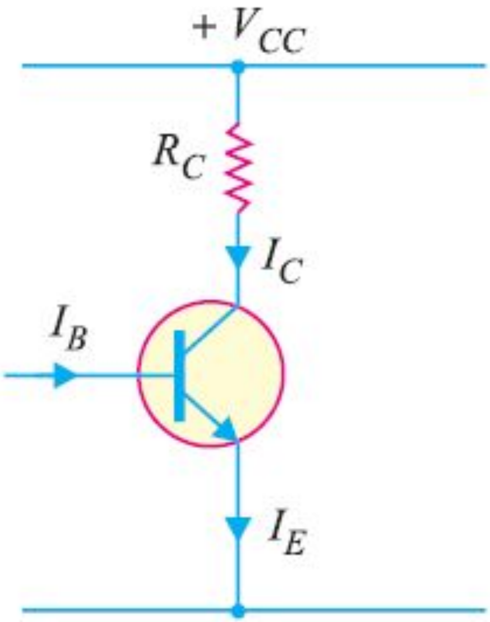
$\therefore$ Zero signal collector current, $I_C = \beta I_B = 50 \times 0.02 = 1$ mA

Zero signal collector-emitter voltage is

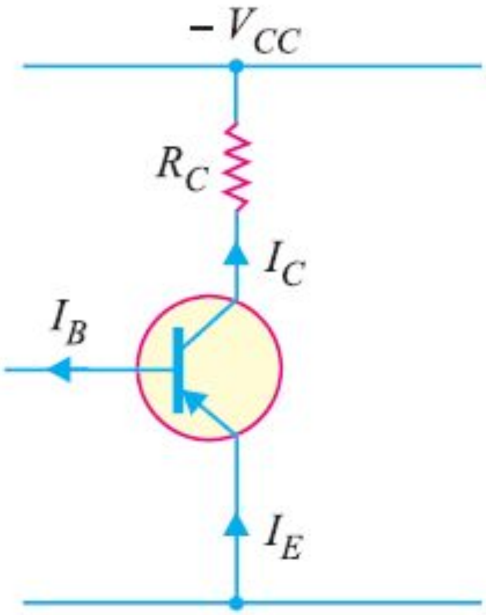
$$V_{CE} = V_{CC} - I_C R_C = 12 - 1\ \text{mA} \times 6\ \text{k}\ \Omega = 6\ \text{V}$$

$\therefore$ Operating point is **6 V, 1 mA**.

Practical Way of Drawing CE Circuit



Common Emitter npn circuit



Common Emitter pnp circuit

Output from Transistor Amplifier

1. First Method: used in single stage amplifier.

$$\text{Output} = \text{voltage across } R_C = i_C R_C$$

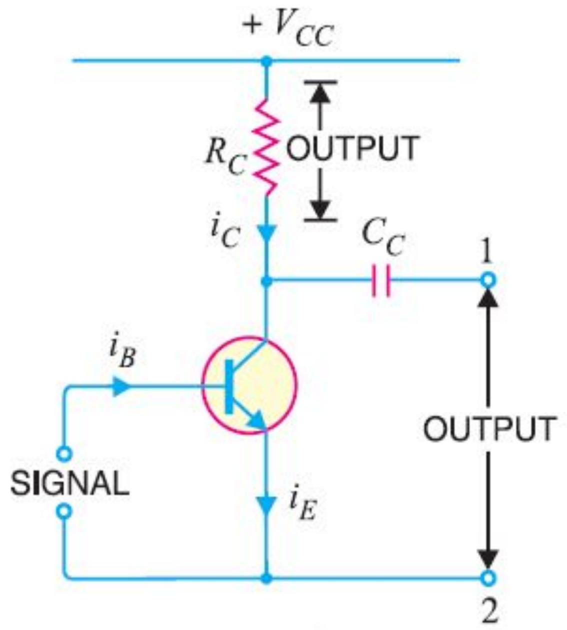
2. Second Method: used in multi-stages amplifier.

$$\text{Output} = \text{Voltage across terminals 1 and 2}$$

$$= V_{CC} - i_C R_C$$

Vcc can not pass through the Capacitor Cc.

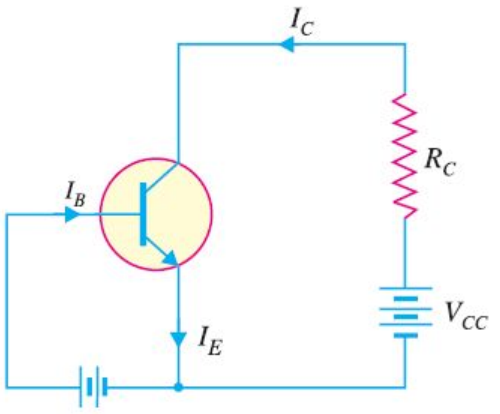
$$Output = -i_C R_C$$

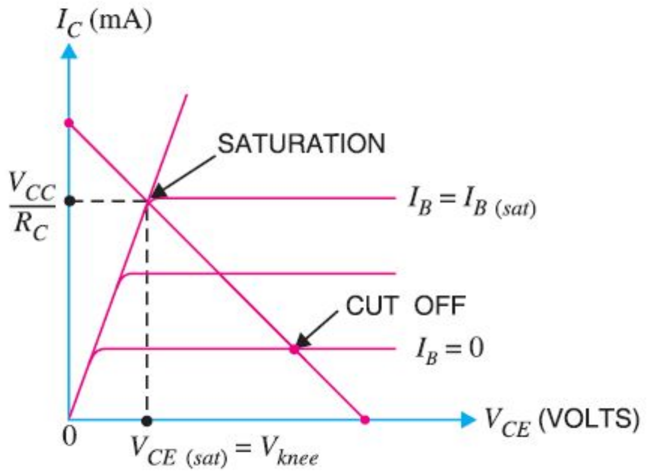


Transistor States

(i) Cut off:

- The point where the load line intersects $I_B = 0$ curve.
- Only leakage current I_{CEO} exits.
- Base-emitter junction no longer remains forward biased.
- Normal transistor action is lost.
- $V_{CE\ (cut\ off)} = V_{CC}$





(ii) Saturation:

- The load line intersects

$I_B = I_{B(sat)}$ curve.

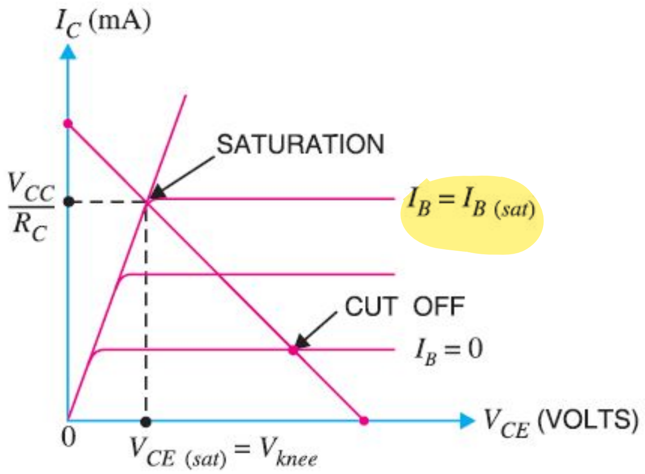
- Base current is maximum

And so is the collector current.

- Collector-base junction no

longer remains reversed biased.

- Normal transistor action is lost.



$$I_{C\,(sat)} \simeq \frac{V_{CC}}{R_C}$$

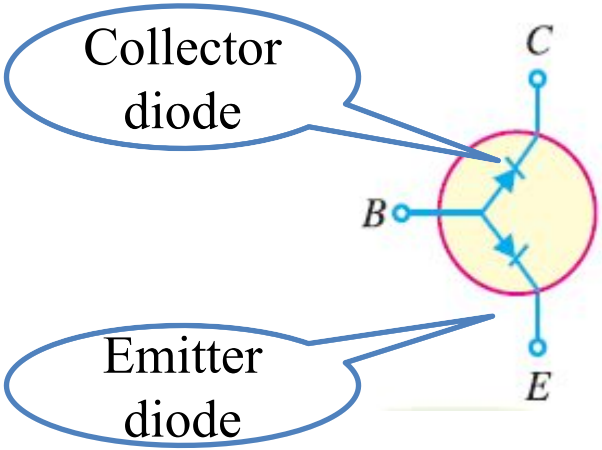
$$V_{CE} = V_{CE(sat)} = V_{knee}$$

(ii) Active Region:

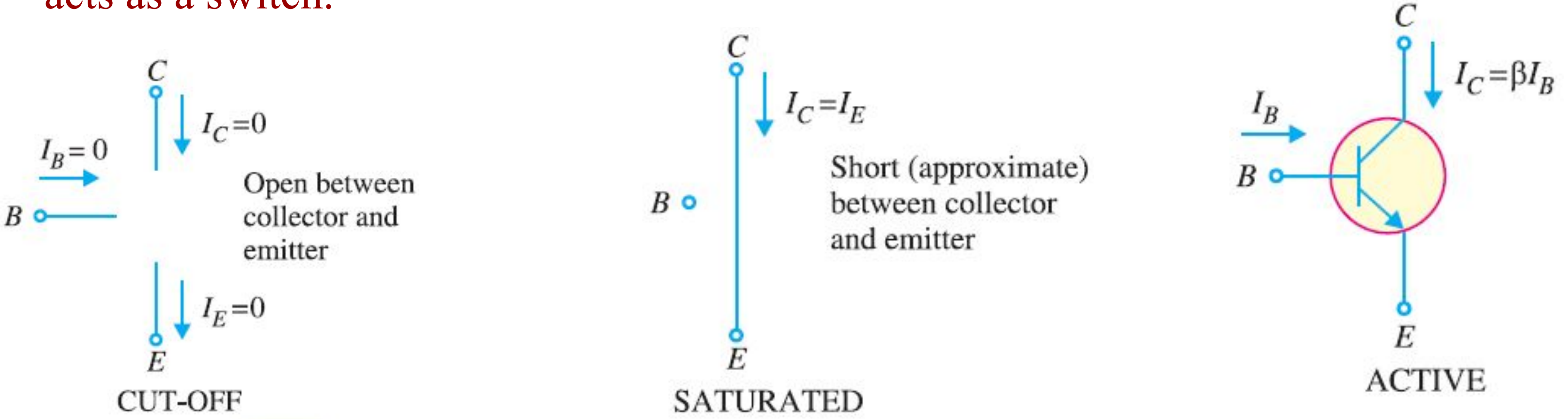
- Region between cut off and saturation.
- Collector-base reverse and base-emitter forward biased.
- Function normally.

Relation between diode and transistor states:

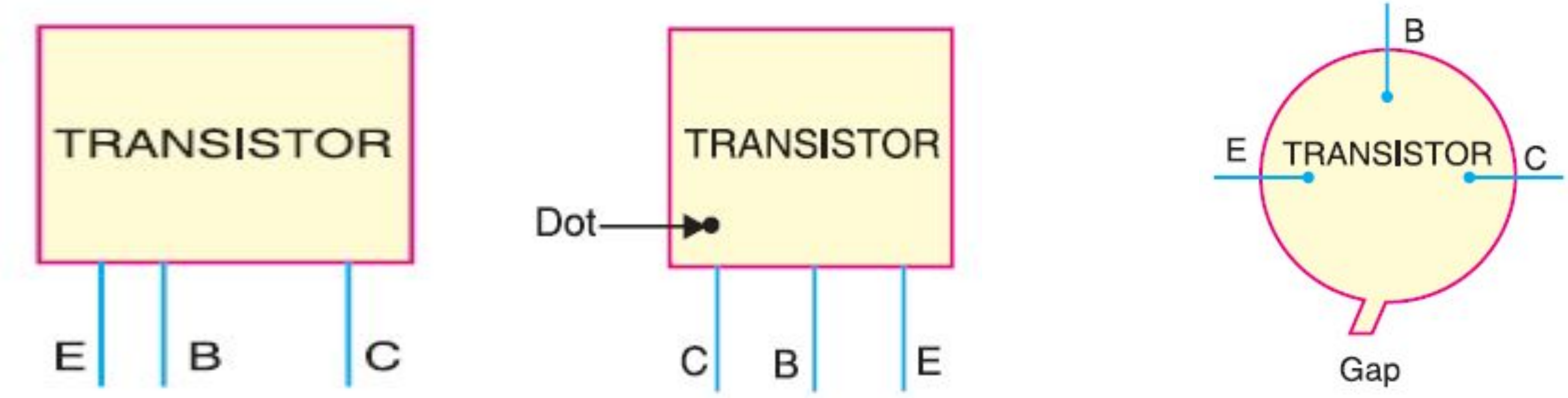
- Cut-off: Emitter diode and collector diode are OFF.
- Saturated: Emitter diode and collector diode are ON.
- Active: Emitter diode is ON and collector diode is OFF.



So in Active region transistor acts as an amplifier and in Cut-off and saturation region it acts as a switch.



Transistor Lead Identification



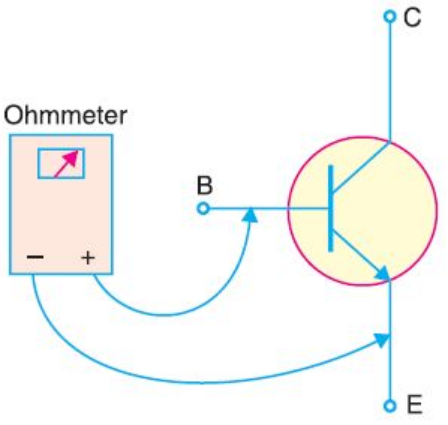
Transistor Testing

For npn transistor:

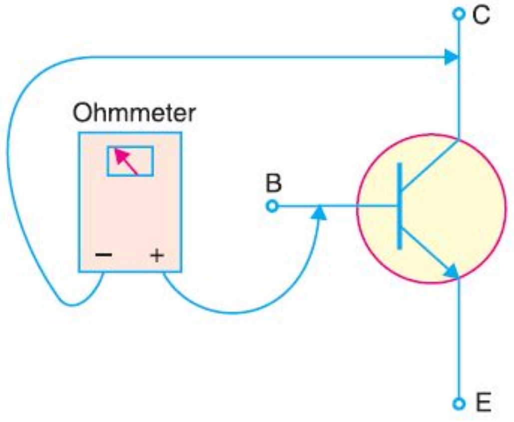
i. The base-emitter resistance is 100 Ω to 1 KΩ. If not, it is faulty.

ii. The collector-base resistance is 100 KΩ or higher. If not, it is faulty.

For pnp transistor: the ohm-meter lead must be reversed.



(i)



(ii)

Transistors Versus Vacuum Tubes

Advantages of Transistors:

- High voltage gain.
- Lower supply voltage.
- No heating.
- Much smaller.
- Mechanically strong due to solid-state.

Disadvantages of Transistors:

- Lower power dissipation.
- Lower input impedance.
- Temperature dependence.
- Inherent variation of parameters.